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# 2 Data Management SS2026: Exercise 02 – Ingestion and Queries

**Published: Wed. 01.04.2026**
**Deadline: Fri. 08.05.2026, 11.59pm**

- This exercise on data ingestion and query languages aims to provide practical experience with the open-source database management system (DBMS) PostgreSQL and the Structured Query Language (SQL).
- We expect you to submit individual files containing the partial results of the sub-tasks on TeachCenter only (**no zip archive**).
- This exercise is subject to automated testing. Any partial submission that fails the testing system due to extraneous statements or similar errors will receive a grade of **0 points**.
- Incorrect file naming will also receive a grade of **0 points**.
- **IMPORTANT:** This exercise must be completed individually. Any form of plagiarism, copying, AI usage, or unauthorized collaboration will be sanctioned according to academic regulations. Strict measures will be taken against any violations.

## 2.1 Database Creation and Data Ingestion (5/30 points)

As a preparation step, set up the DBMS PostgreSQL (free, pre-built packages are available for Windows, Linux, Solaris, BSD, macOS). Then create a new database and set up the schema provided via the following link[1]. Please note that the data structure in this exercise differs from the one described in the previous one. Thus, under no circumstances, use the schema you have created for the first exercise.

Next, retrieve the data (Ex2Data.zip) from the following link[2] containing all the necessary `.csv` files. These `.csv` files serve as the data source for this task. Your task is to ingest the data into the SQL tables using the COPY SQL commands, e.g.: "COPY *table-name* FROM *file-name*". More detailed information (especially with respect to the additional parameters you will need) can be found in the lecture slides, and in the official PostgreSQL documentation[3]. In general, please make sure to provide the full file paths, and make sure that there are sufficient read and execute access rights (e.g., in Windows for user "Everyone") to the csv files (and the folder, in which they can be found).

[1] `https://tc.tugraz.at/main/mod/resource/view.php?id=554255`
[2] `https://tc.tugraz.at/main/mod/resource/view.php?id=554249`
[3] `https://www.postgresql.org/docs/current/sql-copy.html`

You are required to write SQL COPY commands to import the data from the `.csv` files into your database tables (one line per table).

**Deliverable Task 2.1:** `IngestData.sql` (File containing SQL commands).

## 2.2 SQL Query Processing (25/30 points)

Having populated the created database in Task 2.1, it is now ready for query processing. Create SQL queries to answer the following questions and tasks. The expected results (Ex2Results.zip) per query is provided via the following link[4]. You can use this to check if your results are correct.

- **Q01 (2 points):** Identify "day-one" converters by retrieving full profiles for customers whose purchases occurred on their corresponding sign-up dates. Return all available customer attributes. Sort the results by customer ID in descending order.
- **Q02 (3 points):** Identify countries with a higher-than-average volume of discounted sales. Return the country names along with their respective total discounted sales figures. Sort the results by total discounted sales in descending order.
- **Q03 (5 points):** Locate the dead stock by retrieving products that are currently in inventory within a specific country but have recorded zero sales in it. Return the product ID and name as well as the associated country from the stock records. Sort the results alphabetically by country, then by product ID in ascending order.
- **Q04 (7 points):** Identify the best performing products within each category by selecting items in the upper quartile (top 25%) based on the category-specific sales volume. Sales volume is defined as the total units sold per product. Return the category, product ID and name as well as the total sold quantity of these top-tier items. Sort the results in the output alphabetically by category, then by product ID in ascending order.
- **Q05 (8 points):** List the top 15 customers who were most active during the business's historical peak sales month by retrieving their purchase frequency and total item volume. The "peak sales month" is the calendar month with the highest count of unique transactions; "purchase frequency" is the count of distinct sales, and "total item volume" is the sum of all units purchased. Return the customer ID, country, purchase frequency, and total item volume for that month. Sort the results by total item volume (descending), then by purchase frequency (descending), and finally by customer ID (ascending) to resolve ties.

**Deliverables Task 2.2:** `Q01.sql`, `Q02.sql` ..., `Q05.sql`. (Five SQL scripts, one for each query)

[4] `https://tc.tugraz.at/main/mod/resource/view.php?id=554252`